

Computer Architecture

D5_Exercise

Consider the following truth table

<i>a</i>	<i>b</i>	<i>c</i>	<i>f₁</i>
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

Question 01

- Obtain the SOP expression.
- Implement the logic circuit.
- Simplify the SOP expression using Karnaugh map.
- Implement the logic circuit of simplified sop expression.

Question 02

- Obtain the POS expression.

- b) Implement the logic circuit.
- c) Simplify the POS expression using Karnaugh map.
- d) Implement the logic circuit of simplified sop expression.